

Meta-analysis handover

Intervention meta-analysis · 12 randomised trials · 9,014 participants · risk ratio · PRISMA 2020 · RoB 2 · GRADE

This is a redacted sample. It shows the structure and level of detail of a real deliverable. Client names, journal names and identifying particulars have been removed or changed. Ask us and we will send one built around an analysis like yours.

1 Statistical analysis plan, as registered

Written and registered before any pooled estimate was produced. Everything below was fixed at that point, which is what allows the methods section to describe these analyses as pre-specified and to be believed when it does.

Element	As pre-specified
Effect measure	Risk ratio with 95% confidence intervals
Primary model	Random effects, DerSimonian–Laird estimation of between-study variance
Sensitivity model	Fixed effect, inverse variance
Heterogeneity	Cochran’s Q, I ² and τ^2 , with a prediction interval reported alongside
Subgroups	Timing of intervention; trial size; risk-of-bias category. Compared by test for interaction, not by significance within each group
Sensitivity analyses	High risk of bias excluded; leave-one-out; fixed-effect model
Small-study effects	Funnel plot with Egger’s regression test, since ≥ 10 trials contribute
Software	R 4.4 with meta 7.0 and metafor 4.6; pooled estimates cross-checked in RevMan
Certainty	GRADE, per outcome, in GRADEpro GDT

2 Included trials and effect sizes

Effect sizes were recomputed from the reported event counts rather than transcribed from abstracts or from another review’s table. Extraction was done in duplicate and reconciled.

Trial	Intervention	Control	Risk ratio [95% CI]	Weight
01 · Netherlands, 2012	24/420	52/418	0.46 [0.29, 0.73]	8.9%
02 · Italy, 2014	23/214	27/210	0.84 [0.50, 1.41]	7.6%
03 · United Kingdom, 2016	61/712	84/706	0.72 [0.53, 0.98]	13.6%
04 · Egypt, 2015	5/132	16/128	0.30 [0.11, 0.80]	2.8%
05 · Japan, 2018	55/560	62/556	0.88 [0.62, 1.24]	12.5%

Trial	Intervention	Control	Risk ratio [95% CI]	Weight
06 · Brazil, 2013	16/268	33/265	0.48 [0.27, 0.85]	6.7%
07 · India, 2011	17/180	18/178	0.93 [0.50, 1.75]	5.8%
08 · Germany, 2020	71/905	101/898	0.70 [0.52, 0.93]	14.6%
09 · Turkey, 2017	10/196	25/192	0.39 [0.19, 0.79]	4.9%
10 · Spain, 2019	34/340	37/338	0.91 [0.59, 1.42]	9.5%
11 · China, 2010	4/104	14/102	0.28 [0.10, 0.82]	2.4%
12 · Canada, 2021	36/498	57/494	0.63 [0.42, 0.93]	10.7%

Weights are from the random-effects model. Risk of bias: trials 07 and 11 rated high, trials 04, 06 and 09 some concerns, the remainder low.

3 Results

Analysis	Trials	Risk ratio [95% CI]	Note
Primary — random effects	12	0.66 [0.55, 0.78]	Pre-specified primary analysis
Sensitivity — fixed effect	12	0.68 [0.60, 0.77]	Agrees with the primary model
High risk of bias excluded	10	0.66 [0.56, 0.79]	Conclusion survives
Subgroup — within 24 h	6	0.48 [0.36, 0.65]	Test for interaction $p = 0.009$
Subgroup — after 24 h	6	0.76 [0.65, 0.88]	Difference between groups, not two results

Heterogeneity

I^2 37%, τ^2 0.033, Q 17.6 on 11 degrees of freedom. Moderate, and largely accounted for by the timing subgroup rather than left in the results as an unexplained number. The prediction interval is 0.42 to 1.03 — wider than the confidence interval, and reported alongside it because it answers the question a clinician actually asks: what effect should a new centre expect, rather than how precisely has the average been estimated.

4 Diagnostics

Leave-one-out

No single trial moves the pooled estimate outside 0.63 to 0.69. The result does not rest on one study, which is the most common fragility in an analysis of this size and the first thing a statistical reviewer tests.

Small-study effects

The funnel plot is asymmetric and Egger's regression intercept gives $p = 0.046$. This is reported, and it is not reported as publication bias. The smaller trials in this set are disproportionately the early-administration ones, and the pre-specified timing subgroup already accounts for a difference of the size observed. Both readings are set out in the discussion and the reader is left able to weigh them; asserting either one would be going beyond what the data can settle.

Why this matters. Funnel asymmetry is routinely reported as evidence of publication bias when it can equally reflect a genuine difference between small and large trials. Distinguishing the two is a matter of judgement about the studies, not a test result, and it is the kind of judgement a reviewer notices the absence of.

Data handling queried with the author

- Two trials reported medians with interquartile ranges. Means and standard deviations were derived using the method of Wan and colleagues, the derivation recorded in the extraction file, and a sensitivity analysis run without those trials.
- One trial reported an outcome that was not in its registration and omitted one that was. It was included, downgraded for selective reporting under RoB 2, and named in the discussion.
- One trial reported the primary outcome only in a figure. Counts were requested from the corresponding author, supplied, and the provenance recorded rather than presented as published data.

5 Certainty of evidence (GRADE)

Outcome	Trials	Certainty	Reason for downgrade
Postoperative infection	12	Moderate	Risk of bias — two trials at high risk
Length of stay	9	Moderate	Risk of bias
All-cause mortality	7	Low	Risk of bias; imprecision — interval spans benefit and harm
Serious adverse events	4	Very low	Not pooled — no shared definition; reported under SWiM

6 What is handed over

- The manuscript in IMRaD with a PRISMA 2020-compliant abstract, formatted to the target journal.
- Forest, funnel, subgroup and leave-one-out plots as vector files at journal resolution.
- Study characteristics, risk-of-bias and GRADE summary-of-findings tables.
- The extraction spreadsheet, with the source of every derived number recorded.
- Commented R scripts and the RevMan project, so every figure above can be reproduced.
- The completed PRISMA 2020 checklist and flow diagram.
- The registered analysis plan, with a written note of every deviation from it.

Pubrica supports the work; authorship remains yours. Where our contribution should be acknowledged under ICMJE, the handover drafts the wording for you to approve. Any AI-assisted screening or drafting is declared in the methods.

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